

AI Engineering

Building Applications with Foundation Models

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can also be reused later for training data annotation, as discussed in [Chapter 8](#).

Tie evaluation metrics to business metrics

Within a business, an application must serve a business goal. The application's metrics must be considered in the context of the business problem it's built to solve.

For example, if your customer support chatbot's factual consistency is 80%, what does it mean for the business? For example, this level of factual consistency might make the chatbot unusable for questions about billing but good enough for queries about product recommendations or general customer feedback. Ideally, you want to map evaluation metrics to business metrics, to something that looks like this:

- Factual consistency of 80%: we can automate 30% of customer support requests.
- Factual consistency of 90%: we can automate 50%.
- Factual consistency of 98%: we can automate 90%.

Understanding the impact of evaluation metrics on business metrics is helpful for planning. If you know how much gain you can get from improving a certain metric, you might have more confidence to invest resources into improving that metric.

It's also helpful to determine the usefulness threshold: what scores must an application achieve for it to be useful? For example, you might determine that your chatbot's factual consistency score must be at least 50% for it to be useful. Anything below this makes it unusable even for general customer requests.

Before developing AI evaluation metrics, it's crucial to first understand the business metrics you're targeting. Many applications focus on *stickiness* metrics, such as daily, weekly, or monthly active users (DAU, WAU, MAU). Others prioritize *engagement* metrics, like the number of conversations a user initiates per month or the duration of each visit—the longer a user stays on the app, the less likely they are to leave. Choosing which metrics to prioritize can feel like balancing profits with social responsibility. While an emphasis on stickiness and engagement metrics can lead to higher revenues, it may also cause a product to prioritize addictive features or extreme content, which can be detrimental to users.

Step 3. Define Evaluation Methods and Data

Now that you've developed your criteria and scoring rubrics, let's define what methods and data you want to use to evaluate your application.